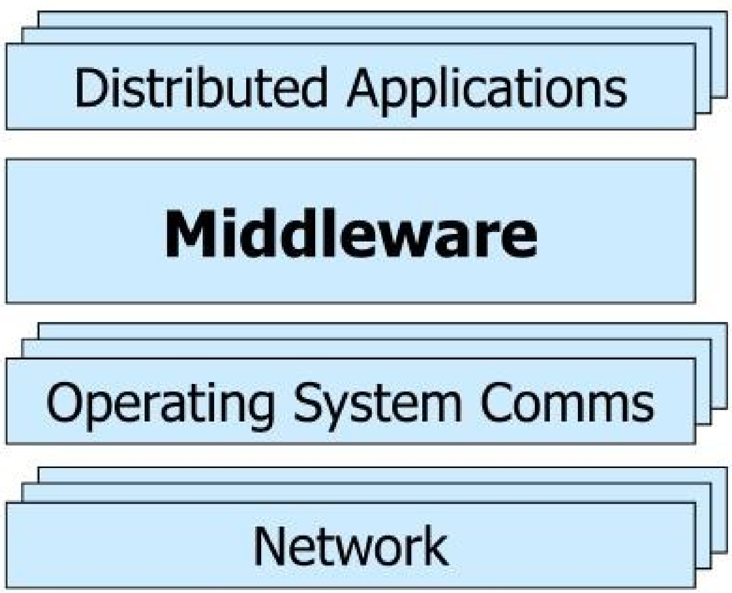
Introduction to Middleware I

- What is Middleware?
  - Layer between OS and distributed applications
  - Hides complexity and heterogeneity of distributed system
  - Bridges gap between low-level OS comms and programming language abstractions
  - Provides common programming abstraction and infrastructure for distributed applications



(remote calls, object invocation, messages, …)

(sockets, IP, TCP, UDP, …)

(packets, bits…)

Introduction to Middleware II

- Middleware provides support for (some of):
  - Naming, Location, Service discovery, Replication
  - Protocol handling, Communication faults, QoS
  - Synchronisation, Concurrency, Transactions, Storage
  - Access control, Authentication

- Middleware dimensions:

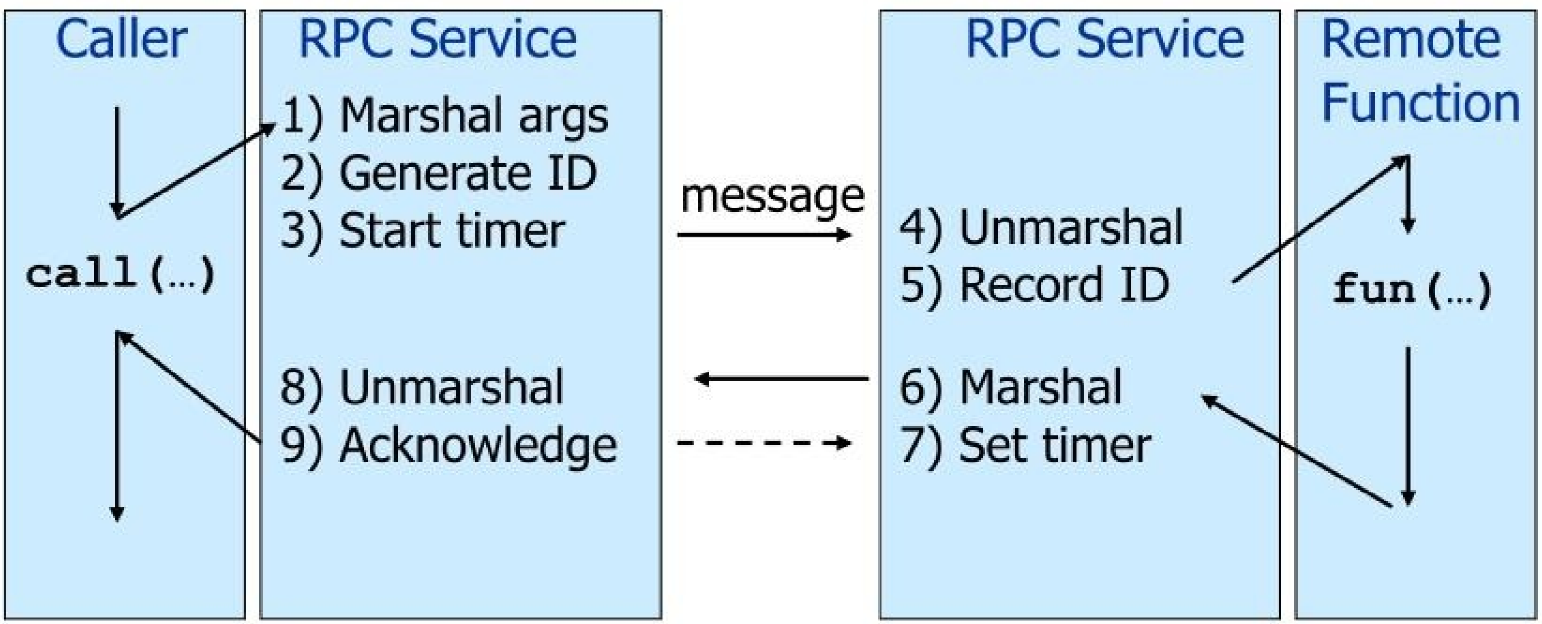
– Request/Reply	vs.	Asynchronous Messaging
– Language-specific	vs.	Language-independent
– Proprietary	vs.	Standards-based
– Small-scale	vs.	Large-scale
– Tightly-coupled	vs.	Loosely-coupled components

Outline

- Part I: Remote Procedure Call (RPC)
  - Historic interest
- Part II: Object-Oriented Middleware (OOM)
  - Java RMI
  - CORBA
  - Reflective Middleware
- Part III: Message-Oriented Middleware (MOM)
  - Java Message Service
  - IBM MQSeries
  - Web Services
- Part IV: Event-Based Middleware
  - Cambridge Event Architecture
  - Hermes

Part I: Remote Procedure Call (RPC)

- Masks remote function calls as being local
- Client/server model
- Request/reply paradigm usually implemented with message passing in RPC service
- Marshalling of function parameters and return value



Properties of RPC

- ✓ Language-level pattern of **function call**
  - easy to understand for programmer
- ✓ **Synchronous request/reply** interaction
  - natural from a programming language point-of-view
  - matches replies to requests
  - built in synchronisation
- ✓ **Distribution transparency** (no-failure case)
  - hides the complexity of a distributed system
- ✓ Various **reliability** guarantees
  - deals with some distributed systems aspects of failure

Failure Modes of RPC

- Invocation semantics supported by RPC in the light of:
  - network and/or server congestion,
  - client, network and/or server failure
    - note DS independent failure modes
- RPC systems differ, many examples, local was Mayflower

Maybe or at most once (RPC system tries once)

- Error return – programmer may retry

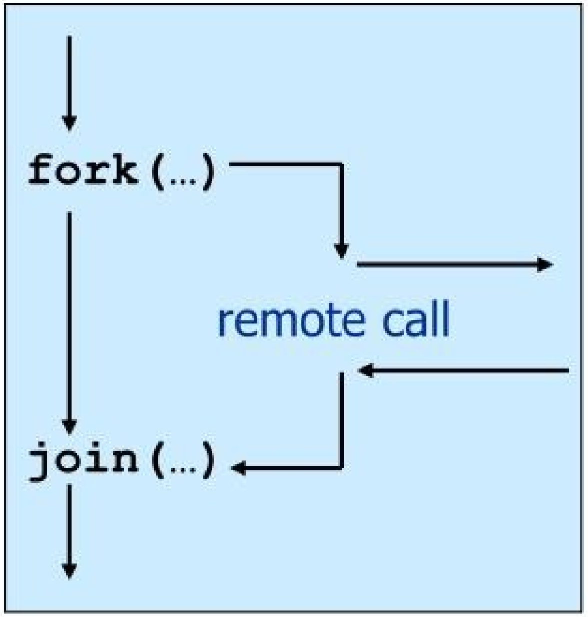
Exactly once (RPC system retries a few times)

- Hard error return – some failure most likely

  note that "exactly once" cannot be guaranteed

Disadvantages of RPC

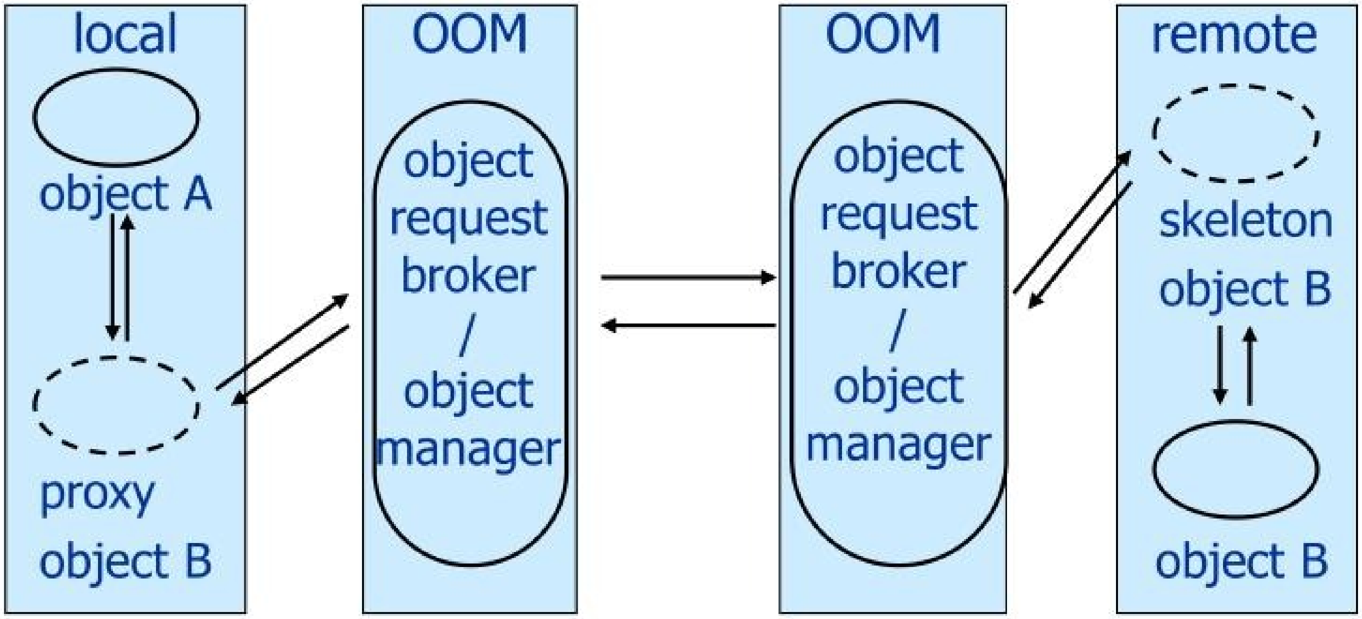
- ✖ Synchronous request/reply interaction
  - tight coupling between client and server
  - may block for a long time
  - leads to multi-threaded programming



- ✖ Distribution Transparency
  - Not possible to mask all problems
- ✖ Lacks notion of services
  - programmer not interested in server but in service
- ✖ RPC paradigm is not object-oriented
  - invoke functions on servers as opposed to methods on objects

Part II: Object-Oriented Middleware (OOM)

- **Objects** can be *local* or *remote*
- **Object references** can be *local* or *remote*
- Remote objects have visible **remote interfaces**
- Masks remote objects as being local using **proxy objects**
- **Remote method invocation**



Properties of OOM

- ✓ Support for object-oriented programming model
  - objects, methods, interfaces, encapsulation, …
  - exceptions (also in some RPC systems e.g. Mayflower)
- ✓ **Location Transparency**
  - mapping object references to locations
- ✓ Synchronous request/reply interaction
  - same as RPC
- ✓ Services

Java Remote Method Invocation (RMI)

- Covered in 1B Concurrent Systems
- Distributed objects in Java

```
public interface PrintServer extends Remote {
  int print(Vector printJob) throws RemoteException;
}
```

- RMI compiler creates proxies and skeletons
- RMI registry used for interface lookup
- All system written in Java (single-language system)

CORBA

- **C**ommon **O**bject **R**equest **B**roker **A**rchitecture
  - Open standard by the OMG (Version 3.0)
  - Language- and platform independent
- **Object Request Broker** (ORB)
  - General Inter-ORB Protocol (GIOP) for communication
  - Interoperable Object References (IOR) contain object location
  - CORBA **Interface Definition Language** (IDL)
    - Stubs (proxies) and skeletons created by IDL compiler
  - Dynamic remote method invocation
- **Interface Repository**
  - Querying existing remote interfaces
- **Implementation Repository**
  - Activating remote objects on demand

CORBA IDL

- Definition of language-independent remote interfaces
  - **Language mappings** to C++, Java, Smalltalk, …
  - Translation by IDL compiler
- Type system
  - *basic types*: long (32 bit), long long (64 bit), short, float, char, boolean, octet, any, …

```
typedef sequence<string> Files;
interface PrintServer : Server {
  void print(in Files printJob);
};
```

  - *constructed types*: struct, union, sequence, array, enum
  - *objects* (common super type `Object`)
- Parameter passing
  - `in`, `out`, `inout`
  - basic & constructed types passed by value
  - objects passed by reference

CORBA Services (selection)

- Naming Service
  - Names → remote object references
- Trading Service
  - Attributes (properties) → remote object references
- Persistent Object Service
  - Implementation of persistent CORBA objects
- Transaction Service
  - Making object invocation part of transactions
- Event Service and Notification Service
  - Asynchronous communication based on messaging (cf. MOM), not integrated programming model with general IDL messages

Disadvantages of OOM

- ✖ Synchronous request/reply interaction only
  - So CORBA `oneway` semantics added and -
  - Asynchronous Method Invocation (AMI)
  - But implementations may not be loosely coupled
- ✖ Distributed garbage collection
  - Releasing memory for unused remote objects
- ✖ OOM rather static and heavy-weight
  - Bad for ubiquitous systems and embedded devices

Reflective Middleware

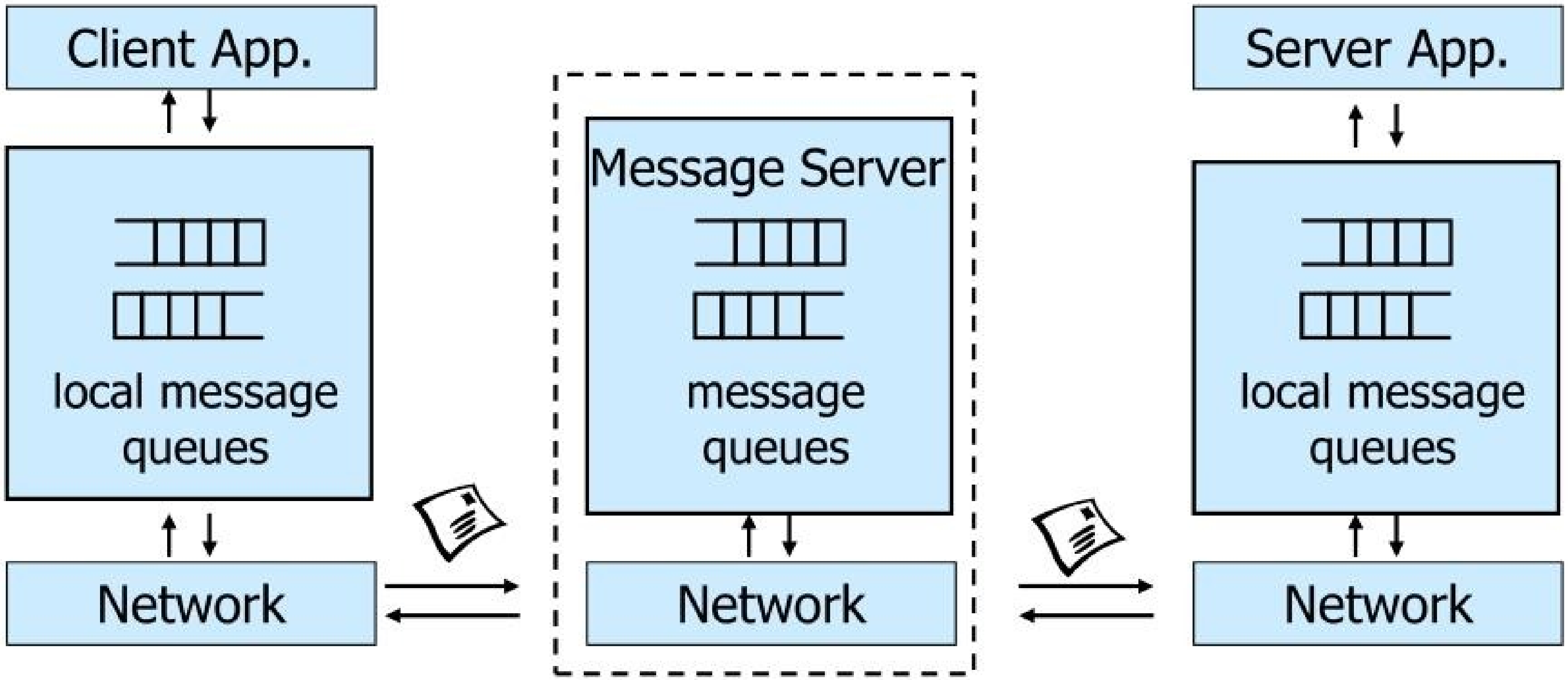
- Flexible middleware (OOM) for mobile and context-aware applications – adaptation to context through monitoring and substitution of components

- Interfaces for **reflection**
  - Objects can inspect middleware behaviour

- Interfaces for **customisability**
  - Dynamic reconfiguration depending on environment
  - Different protocols, QoS, ...
  - e.g. use different marshalling strategy over unreliable wireless link

Part III: Message-Oriented Middleware (MOM)

- Communication using **messages**
- Messages stored in **message queues**
- Optional **message server** decouples client and server
- Various assumptions about **message content**



Properties of MOM

- ✓ **Asynchronous** interaction
  - Client and server are only **loosely coupled**
  - Messages are queued
  - Good for application integration
- ✓ Support for **reliable** delivery service
  - Keep queues in persistent storage
- ✓ Processing of messages by intermediate message server
  - Filtering, transforming, logging, …
  - Networks of message servers
- ✓ Natural for database integration

IBM MQSeries

- One-to-one reliable message passing using queues
  - Persistent and non-persistent messages
  - Message priorities, message notification
- **Queue Managers**
  - Responsible for queues
  - Transfer messages from input to output queues
  - Keep routing tables
- **Message Channels**
  - Reliable connections between queue managers
- Messaging API:

MQopen	Open a queue
MQclose	Close a queue
MQput	Put message into opened queue
MQget	Get message from local queue

Java Message Service (JMS)

- API **specification** to access MOM implementations
- Two modes of operation specified:
  - **Point-to-point**
    - One-to-one communication using queues
  - **Publish/Subscribe**
    - cf. Event-Based Middleware

- **JMS Server** implements JMS API
- JMS Clients connect to JMS servers
- Java objects can be serialised to JMS messages

Disadvantages of MOM

- ✖ Poor programming abstraction (but has evolved)
  - Rather low-level (cf. Packets)
  - Request/reply more difficult to achieve
  - Results in multi-threaded code
- ✖ Message formats unknown to middleware
  - No type checking (JMS addresses this – implementation?)
- ✖ Queue abstraction only gives one-to-one communication
  - Limits scalability (JMS pub/sub – implementation?)

Web Services

- Use well-known web standards for distributed computing

Communication

- Message content expressed in **XML**
- **S**imple **O**bject **A**ccess **P**rotocol (**SOAP**)
  - Lightweight protocol for sync/async communication

Service Description

- **W**eb **S**ervices **D**escription **L**anguage (**WSDL**)
  - Interface description for web services

Service Discovery

- **U**niversal **D**escription **D**iscovery and **I**ntegration (**UDDI**)
  - Directory with web service description in WSDL

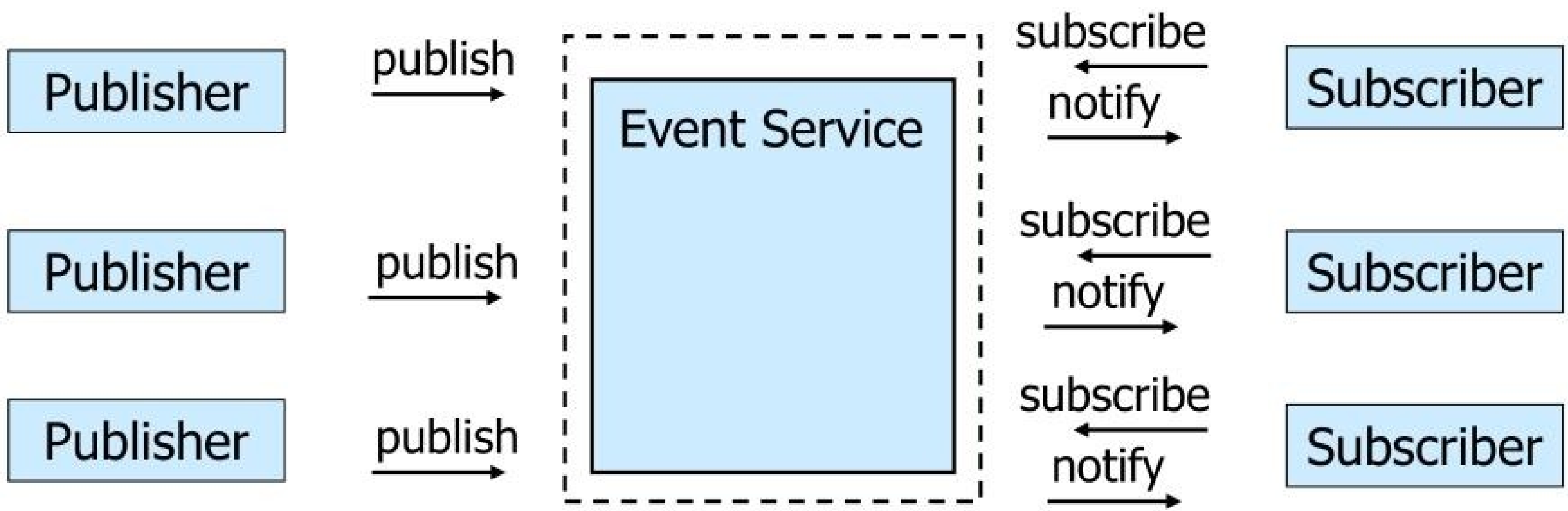
Properties of Web Services

- ✓ Language-independent and open standard
- ✓ **SOAP** is best of both worlds:
  - Synchronous request/reply like OOM
  - Asynchronous messaging like MOM
  - Supports internet transports (http, smtp, ...)
  - Uses XML Schema for marshalling types to/from programming language types
- ✓ **WSDL** says how to use a web service

  http://api.google.com/GoogleSearch.wsdl
- ✓ **UDDI** helps to find the right web service
  - Exports SOAP API for access

Part IV: Event-Based Middleware aka Publish/Subscribe

- **Publishers** *publish* **events** (messages)
- **Subscribers** express interest in events with *subscriptions*
- **Event Service** *notifies* interested subscribers of published events
- Events can have arbitrary content or name/value pairs



Topic-Based and Content-Based Pub/Sub

- Event Service matches events against subscriptions
- What do subscriptions look like?

Topic-Based Publish/Subscribe

- Publishers publish events belonging to **topic** or **subject**
- Subscribers subscribe to **topic**

```
subscribe(PrintJobFinishedTopic, …)
```

(Topic and) Content-Based Publish/Subscribe

- Publishers publish events belonging to **topic**s and
- Subscribers provide a **filter** based on ***content*** of events

```
subscribe(type=printjobfinshed, printer='aspen', …)
```

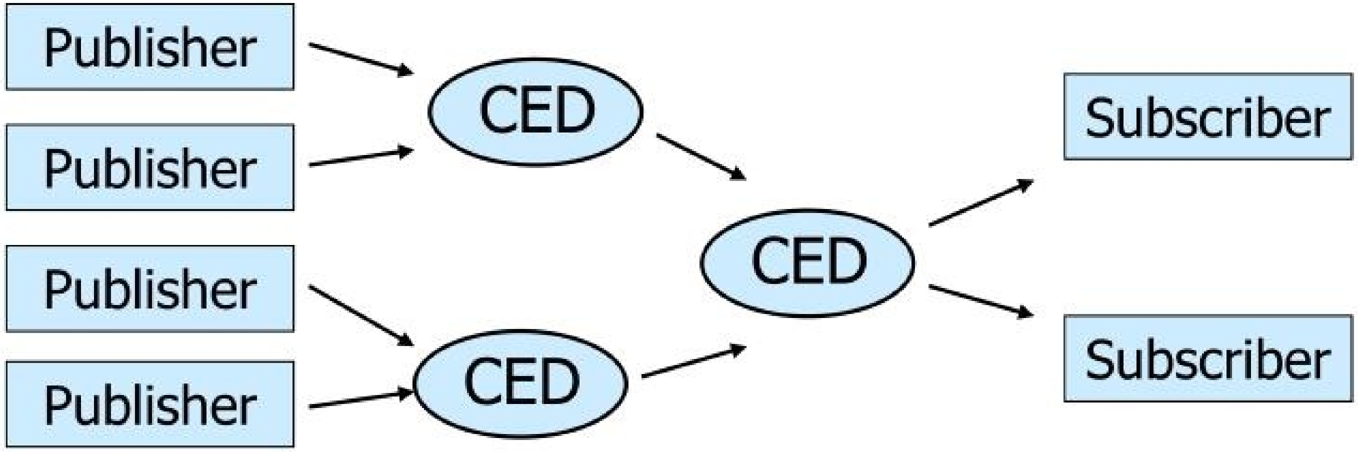
Properties of Publish/Subscribe

- ✓ Asynchronous communication
  - ✓ Publishers and subscribers are loosely coupled
- ✓ Many-to-many interaction between publishers and subscribers
  - ✓ Scalable scheme for large-scale systems
  - ✓ Publishers do not need to know subscribers
- ✓ (Topic and) Content-based pub/sub very expressive
  - ✓ Filtered information delivered only to interested parties

Composite Event Detection (CED)

- Content-based pub/sub not expressive enough
  - Potentially thousands of event types
  - Subscribers interested in *patterns of events*
- **Event Patterns**

  `PrinterOutOfPaperEvent` or `PrinterOutOfTonerEvent`
- **C**omposite **E**vent **D**etectors (CED)
  - Subscribe to primitive events and publish composite events



Summary

- Middleware is an important abstraction for building distributed systems

1. Remote Procedure Call
2. Object-Oriented Middleware
3. Message-Oriented Middleware
4. Event-Based Middleware

- Synchronous vs. asynchronous communication
- Scalability, many-to-many communication
- Language integration
- Ubiquitous systems, mobile systems